

LLD Home Notes - Index

Each DESIGN.pdf = how-it-works notes + full DESIGN.md + project source code. Print double-sided to save paper.

- 1. Parking Lot - How it works -> lld/parking_lot/DESIGN.pdf (29 code files)

Car comes in -> find a free spot -> give ticket -> on exit charge money -> free the spot.

- 2. ATM - How it works -> lld/atm/DESIGN.pdf (18 code files)

Insert card -> PIN -> withdraw -> machine gives notes -> card comes out.

- 3. Car Rental - How it works -> lld/car_rental/DESIGN.pdf (29 code files)

Pick branch + car type -> system chooses a free car -> price it -> pay -> later return at drop branch.

- 4. Splitwise - How it works -> lld/splitwise/DESIGN.pdf (17 code files)

Friends share expenses -> system tracks who owes whom -> simplify into fewer payments.

- 5. Doctor Appointment - How it works -> lld/doctors_appointment/DESIGN.pdf (19 code files)

Doctor opens slots -> patient searches -> books -> if taken, waitlist -> cancel promotes next patient.

- 6. Customer Support - How it works -> lld/customer_support/DESIGN.pdf (12 code files)

Customer raises issue -> assign to free agent with right skill -> agent resolves -> history saved.

- 7. Logger - How it works -> lld/logger/DESIGN.pdf (17 code files)

You call logger.error('msg') -> message walks a chain of level handlers -> subscribed outputs print/write it.

- 8. Rate Limiter - How it works -> lld/rate_limiter/DESIGN.pdf (11 code files)

Before handling an API call, ask: has this user used too many requests? Allow or block.

- 9. Snakes & Ladders - How it works -> lld/snakes_and_ladder/DESIGN.pdf (16 code files)

Players take turns rolling dice, move on a serpentine board, snakes send you down, ladders up, exact land to win.

- 10. Custom HashMap - How it works -> lld/hashmap/DESIGN.pdf (3 code files)

Keys go into buckets by hash. Collisions chain in a list. Too full? Grow and rehash.